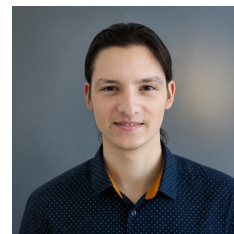


Bruno Ferreira



Applied AI Engineer

Python · LLMs · RAG · Document automation

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brunocar.ferreira@gmail.com · [LinkedIn](#) · [Portfolio](#)

I build AI applications that turn complex documents into structured data and review workflows. My experience combines client engineering in LegalTech, technical specifications and quality validation with 4+ years across AI research, university teaching and applied projects.

TECHNICAL SKILLS

AI engineering: LLMs, RAG, embeddings, NLP, OCR, information extraction, evaluation

Software: Python, FastAPI, TypeScript, React, Next.js, SQL, PostgreSQL, Docker, Git

Delivery: Process discovery, requirements, API integration, testing, documentation, training

PROFESSIONAL EXPERIENCE

Applied AI Engineer and Consultant

Independent client work · Portugal and remote

Jun 2025–present

- Own technical delivery within each client project scope: workflow analysis, solution architecture, implementation, testing and handover documentation.
- Advise on AI use cases and acceptance criteria; design and deliver practical training for client and developer teams.

SELECTED CLIENT PROJECTS

LegalTech platform for US visa workflows

Components in development

2025–present

- **Structured intake and document workflows.** Designed and implemented triage, dynamic questionnaires and document-management components in a multi-tenant application.
- **Evidence-guided reports.** Combined OCR and LLM analysis to support legal review against USCIS criteria.

Python, FastAPI, Next.js, PostgreSQL, Redis, Alembic, Docker, OCR, LLMs

HVAC specification extraction

Technical project

Jan–Mar 2026

- **PDF specifications to Excel and JSON.** Built a Python extraction pipeline for engineering review, with source-page evidence and evaluation against manual references.
- **Hybrid extraction.** Combined embedding-based RAG, deterministic table mapping and regular expressions with vision and language models.

Python, RAG, embeddings, PDF/table extraction, vision, LLMs, Excel, JSON

Quality Copilot for packaging and technical documents

Private pilot

2026

- **Document-validation workflows.** Built modules for dimensional checks, content validation, correction verification and technical-sheet preparation.
- **Reports for human review.** Defined the evidence required at each stage and generated verifiable outputs in a private pilot.

Python, OCR, Claude Code skills, document automation

TEACHING AND RESEARCH

Invited Lecturer

University of Coimbra

Feb 2024–present

- Teach databases, AI, Java, C, natural language processing and knowledge representation.
- Prepare learning materials, assess student projects and supervise MSc dissertations.

Researcher

CISUC · Centre for Responsible AI

Feb 2022–Jun 2025

- Researched NLP, information extraction, question answering and the evaluation of knowledge encoded in language models.
- Published work on language-model evaluation and document information extraction, alongside research in explainable AI, causal inference and model compression.

EDUCATION

PhD in Informatics Engineering

University of Coimbra · In progress

2022–present

MSc in Informatics Engineering — Intelligent Systems

University of Coimbra

2022

BSc in Informatics Engineering

University of Coimbra

2021

ADDITIONAL WORK

Privacy-by-design AI platform

AI and privacy

2026

Built a functional React/TypeScript PWA using local NER and ONNX for personal-data detection and anonymisation before external LLM workflows.

THE 17.3 evidence evaluation

Commissioned project

2025

Designed and carried out a repeatable review of public university evidence across 17 THE Impact Rankings indicators.

SELECTED PUBLICATIONS

[Towards Automated Evaluation of Knowledge Encoded in Large Language Models · DLnLD @ LREC-COLING 2024](#)

[BATS-PT: Assessing Portuguese Masked Language Models in Lexico-Semantic Analogy Solving and Relation Completion · PROPOR 2024](#)

[Leveraging Question Answering for Domain-Agnostic Information Extraction · CIARP 2023](#)

[Full publication list and project details](#)

LANGUAGES AND AVAILABILITY

Portuguese: native · English: advanced · Spanish: good comprehension, basic speaking

Based in Pombal, Portugal. Open to Applied AI / AI Solutions roles and B2B engagements through my company, including defined projects and ongoing contracts.

Development tools: Claude Code, Codex, OpenCode and Pi, with my own code review and testing.